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A PROJECT REPORT
ON

"VIDEO STEGANOGRAPHY"

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CERTIFICATE

This is to certify that Project work Phase II entitled **"VIDEO STEGANOGRAPHY"** is a bonafide work carried out by **BHUVAN K [1JS19IS022], DHANUSH H [1JS19IS027], HARSHA S M [IJS19IS037], SUNIL K M [1JS20IS410]** in partial fulfilment for the award of degree of Bachelor of Engineering in Information Science and Engineering of Visvesvaraya Technological University Belagavi during the year 2022-2023.

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ABSTRACT

The process that data is communicated has modified significantly as a result of the Internet and computer technology. This has allowed for the use of steganography in an entirely new way for guaranteeing secure data transport. A sophisticated way to hide information is using steganography. By concealing the message in the carrier file, it is feasible to completely deny that any message ever existed. In this work, a stego machine is built with the purpose of disguising text-containing data in computer video files and recovering the concealed data. The Least Significant Bit (LSB) alteration method may be used to embed a text file into a video file without impairing the video's usefulness to achieve this. The modification made by this means are modest. This method's goal is strong security, making it impossible for an eavesdropper to find secret information.

The use of digital media such as Text, Image and Video is increasing day by day. This leads to need for security of digital media to prevent unauthorized access. One of the security techniques we can use is Steganography. The process of hiding required data in any type of digital content is known as Steganography. Steganography is derived from a word Steganos, means hidden or covered and root graph indicates to write. In case of Cryptography the data can be hidden is encrypted and was transmitted. In this type of cases the risk of third-party access is high. While in the steganography the encrypted content was hidden in cover source which makes it difficult for third party access.

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LIST OF ABBREVIATIONS

GUI	- Graphical User Interface
LSB	- Least Significant Bit
BMP	- Bitmap
DES	- Data Encryption Standard
Triple DES	- Triple Data Encryption Standard
AVI	- Audio Video Interleave
MPEG	- Moving Picture Experts Group
MP4	- MPEG Part-14
JPEG	- Joint Photographic Experts Group
PNG	- Portable Network Graphic
GIF	- Graphic Interchange Format
MSB	- Most Significant Bit
AWT	- Abstract Window Toolkit

Chapter 1

INTRODUCTION

1.1 Background

It is an age-old skill known as steganography to send messages in a covert manner so that only the recipient is aware they have been got back. In recent months, the news media and a number of intelligence organizations have brought the topic of steganography into the public eye. The agencies are assisting themselves in achieving their goals by utilizing steganography and encryption in addition to cutting-edge communication technologies and media. Thus, imperceptibility—the ability of the implanted signals to be invisible to the human eye—is a basic prerequisite for a steganographic tactics. Greek words steganos, which means hidden or covered, and graphy, which means writing or drawing, are the origins of the English word steganography. It's also known as Stego or steganography. The theory of this technique has been around for an eternity. On wax-coated tablets, the Greeks proceeded to write words to one another. The tablet's wax was at first scraped off, after which its hidden idea was composed on it, the wax was then reestablished.

1.2 Existing Systems and their Limitations

People of all ages are sharing personal information online as the use of computers and the internet grows but now that we live in a digital age experts are seriously considering how to preserve personal data a method called steganography can be used to hide personal information inside cover files so that only the sender and recipient are aware of its presence although steganography does not require a secret key utilizing private keys can increase its security cover files are initially encrypted with a key to prevent unauthorized access to the sensitive data combining encryption and steganography however can make the approach more difficult because it may take longer overall to insert the concealed . Additionally, the cost of the technique may increase due to the expansion of hardware components.

Steganography methods have several significant drawbacks. These issues include restricted capacity to conceal data, inferior graphic quality in stego files, weaker resistance to attacks, compatibility with multiple cover file formats, and concerns regarding authenticity, integrity, confidentiality, and flexibility. However, using video cover file can alleviate some

of these problems due to its unique features. Steganography is categorized into four groups based on the carrier file utilized, which are text, image, audio, and video.

1.2 Proposed System

All of these different kinds of video files are becoming more and more commonplace. Video files are used in a variety of industries, including social media, banking, education, and medicine. Compared to other types of files, such as text, photos, and audio, video files are larger in size. The video's enormous size allows for better concealment. There are many ways to hide sensitive information in a video file. This can enhance the technique's security. The likelihood of discovering secret information is decreased by video's dynamic nature. By doing this, robustness against many forms of attacks can be improved. Concealing ability, imperceptibility, and robustness are some of the main problems with the current steganography techniques.

As a result, research on video data concealing is still ongoing. There are two main approaches to conceal data in video clips.: bitstream-level and data-level. In bitstream-level, the existing compression standards' redundancies are utilized at the bitstream level. Encoders typically provide a variety of possibilities, and this freedom of choice is appropriate for manipulation with the goal of data concealing. These methods, however, heavily rely on the bitstream's structure, making them rather fragile in the sense that they frequently cannot withstand any format conversion or transcoding, even without suffering a large loss in perceptual quality. Therefore, this kind of data masking techniques are typically recommended for delicate applications, like authentication. However, data-level approaches are more resistant to intrusions. They can thus be used for a wider variety of applications. The bitstream based techniques are nevertheless appealing for applications requiring data concealing notwithstanding their vulnerability. For instance, the redundancy in the H.264 encoding's block size selection is used to conceal data.

In this paper, we propose a new block-based selective embedding type data hiding framework that encapsulates LSB and RA codes in accordance with an additional temporal synchronization mechanism. LSB is a practical data hiding method, which is shown to be superior to the conventional Quantization Index Modulation (QIM).

The major contribution of this proposed system is as follows:

- The key attributes of the steganography approach are capacity and imperceptibility. The embedded data must be imperceptibly invisible to the viewer in order to be considered imperceptible.
- Using four separate video streams (Road.avi, India.avi, Birds.avi, and Paint.avi) and one secret text (msg.txt), the effectiveness of the suggested approach is assessed.
- By contrasting the original film with its Stego version, it is possible to determine the contained data's perceptual imperceptibility. There won't be any quality loss when we conceal text in video, and nobody will ever be able to tell that there is data present.
- We discovered that utilising the OpenCV tool and the hash-based least significant bit approach for video steganography increases PSNR.
- More hiding capacity results in higher PSNR and lower MSE since video has a high hiding capacity. It provides various PSNR values that relate to the video quality. This number varies depending on the resolution.

A. Least Significant Bit (LSB) Modification Method

The least significant bit (LSB) approach is used by this stego machine to conceal the data in a video clip. The minimum computational cost and exceptionally large watermark channel bit rate of the LSB coding scheme are its main advantages. The watermark implanted using the LSB coding approach gets more resistant as greater LSB depth is used to conceal data. The process involves modifying the least significant bits of each pixel in the carrier file to encrypt hidden data [6]. Three hidden bits, one for each of the RGB values, might fit into each pixel in the present scenario.

B. Advantages of Proposed System

The projected stego machine's benefits include:

- A wizard-based system GUI (Graphical User Interface) that is very easy to use and attractive.
- The capacity to use the system without any prior instruction or reference to any help files.
- The ability to accurately hide and disclose secret data from video files without interfering with already-running or newly-installed applications.
- Capability of using graphics and data to encrypt and decode data.With this approach, an image won't lose quality after the data has been hidden.

Chapter 2

LITERATURE SURVEY

A literature survey or a literature review in a project report shows the various analyses and research made in the field of interest and the results already published, considering the various parameters of the project and the extent of the project. The primary goal of a literature review is to analyse the project's history in order to identify any weaknesses in the current system and provide recommendations on how to remedy any outstanding issues. Therefore, the following themes not only provide the project's history but also expose the issues and shortcomings that prompted the development of this project.

A literature survey is a text of a scholarly paper, which includes the current knowledge including substantive findings, as well as theoretical and methodological contributions to a particular topic. Literature reviews use secondary sources, and do not report new or original experimental work. Most often associated with academic-oriented literature, such as a thesis, dissertation or a peer- reviewed journal article, a literature review usually precedes the methodology and results sectional though this is not always the case. Literature reviews are also common in are search proposal or prospectus (the document that is approved before a student formally begins a dissertation or thesis). Its main goals are to situate the current study within the body of literature and to provide context for the particular reader. Literature reviews are a basis for researching nearly every academic field. A literature survey includes the following:

- Existing theories about the topic which are accepted universally.
- Books written on the topic, both generic and specific.
- Research done in the field usually in the order of oldest to latest.
- Challenges being faced and on-going work, if available.

Literature survey describes about the existing work on the given project. It deals with the problem associated with the existing system and also gives user a clear knowledge on how to deal with the existing problems and how to provide solution to the existing problems.

[1] Rajalakshmi et al., have developed image processing applications, as security for videos challenging job. Some of the techniques are suggested in the existing work for video security and embedding. Steganography in video is seen as a tool of concealment. The most important issues with secure data transfer in the electronic age were information and secret

communication. This paper's primary goals are to maintain the confidentiality of the communicated data and to ensure its efficient transfer. Sensitive data has always been a key concern, both historically and currently. In order to make it difficult for an attacker to discover the message, an innovative technique is proposed in this research. This study discusses video steganography algorithms that use Patch wise Code Formation techniques to conceal video files inside other videos. This method offers reliability, higher video quality, and authentication capabilities. PSNR, MSE, CR, and BPP indicators are used to gauge the process' efficiency in terms of compression ratio performance.

[2] Panado Gede Pradnya Jaya S.T. et al.'s technique includes a spatial domain that operates on bits rather than cover files statistically. The simplest steganography embedding technique is known as least significant bit. In order to strengthen security, we suggested a way of LSB with increased methodology in this study. Steganography is a type of technology that allows for the conveyance of hidden messages. Text, images, and other files that can be transformed into binary, are all acceptable forms of a secret message. The cover file, which can be an image, sound, or video, must, in general, be larger in size than the hidden message to contain it. To conceal secret messages in a cover file, numerous strategies have been suggested. These methods use spatial domain and operate on bits rather than cover files statistically. Least Significant Bits are well renowned for being the most straightforward. In this study, we suggested an LSB method with an improved technique to boost security. One needs a certain Stego key that describes the position of the message and to reconstruct it, in order to extract hidden messages from cover files. Since the majority of video footage are from security camera recordings, movie footage, or personal video footage, include human figures, we use a people detection algorithm as a stego key.

[3] Kousik Dasgupta1 et al., Video Steganography is the practice of concealing sensitive information or data within videos. This study proposes a hash-based least significant bit (LSB) approach. A spatial domain method where the cover frames' LSBs include hidden information. Eight bits of the secret information are divided into 3,3,2 and accordingly encoded into the RGB pixel values of the cover frames. The LSB bits' insert position is chosen using a hash function. Peak Signal to Noise Ratio (P SNR) in comparison to the original cover video and Mean Square Error (MSE) calculated between the original and steganographic files and averaged over all video frames are two metrics used to evaluate the suggested approach. The findings of the measurement of Image Fidelity (IF)indicate that

the steganographic video file hasn't degraded much. The results show that the suggested method performs better than currently used LSB-based steganography.

[4] R K Agarwal and ishitha the development of solutions to protect online information has been facilitated by web development. There are several ways to safeguard data, but one popular one is to obfuscate it using HTML web pages, where it is buried in the source code. Numerous pieces of personal information on the internet need to be hidden. We provide an approach that combines steganography and cryptography techniques to help protect data by concealing it within the HTML source code of online sites. The secret message is encrypted using RSA encryption. By making very minor changes to the code, the encrypted data is compressed, mapped into Space codes, and embedded beneath the HTML web pages where the intruder cannot detect them because they are not suspicious. The security and sturdiness are improved. The method that is most frequently used for hiding personal information on the internet is HTML steganography. This report offers a suggested plan for enhancing the data's security and sturdiness. approach makes the HTML web document more resilient by reducing the page size.

[5] Mritha Ramalingam et al., Data communication has advanced thanks to computer technology and the Internet. This has made it possible to use steganography in a whole new method to guarantee safe data transit. The art of information concealment is called steganography. Deniability of any message's presence is made possible by burying the message in the carrier file. In order to conceal text-containing data in computer video files and to retrieve the hidden information, this paper constructs a stego machine. This can be created by employing the Least Significant Bit (LSB) modification method to embed a text file inside a video file in a way that preserves the video's functionality. This process makes subtle adjustments. This method's goal is high security, making it impossible for an eavesdropper to find secret information.

[6] Suhad Fakrie et al., realm of communications and computer networks, which can be a solution when a secure connection is not available. An information-hiding technique using codded images is presented in this study. A bit plan was used to resize the secret image into a vector, and an Absolute Moment Block Truncation was also used. Data concealing is one of the information security subfields that is utilized in the Coding (AMBTC) process for color images (cover images). By using two methods known as Mb, Ma, and Bitmap to

express each block's location in binary form, the coding technique is used to reduce the quantity of data. Modifying the bit- map simulates the embedding of hidden information. The logistic map is a chaotic map that adds further complexity to the bit plane permutation method. The proposed method was investigated and shown to be effective by obtaining high PSNR and low RMSE results, as well as the visual testing by the histogram test of the image before and after embedding.

[7] W. Zhang et.al Recently, a number of proposed picture encryption techniques have been made, some of them are based on bit-level or pixel-level encryption. However, there are downsides to both bit level and pixel level permutation. In this paper, a novel cryptosystem is proposed as a solution to these problems. First, different permutation methods are carefully evaluated and compared. The Chen system is used to provide a random visiting mechanism to the bit level of the plain-image in a new 3D bit matrix permutation. This is due to the fact that an image may be seen from the bit level as a natural three-dimensional (3D) bit matrix with width, height, and bit length. By leveraging aspects of the Chen system with a 3D Cat map in the permutation stage, a new mapping method is developed to transfer one random location to another random position (double random position permutation) in the 3D matrix as opposed to utilizing traditional sequential visits to the plain-image. We perform simulations, and conclusions shows the usefulness and privacy of our new cryptosystem.

[8] Sghaier Guizni et.al The research community has noticed a rise in wanting about information concealed in multimedia data as the host in recent years. These hidden details may be used for a variety of things, notably source identification, copyright protection, and covert data transmission. In this paper, it is recommended to encrypt and decode audio/video sequences utilising an optical crypto technique with adaptive steganography (AS). Using the optical cryptography technique, which is based on the double random phase encoding method, the desired audio/video sequences are encoded and decoded. Steganography algorithms' primary aim is to bury as much data as they can under the cover material. Therefore, the trade-off for steganography methods is the quantity of hidden data, or "stego-data," that is put in.

[9] **D. Xu et.al** suggested that in order to retain security and privacy, digital images may occasionally need to be saved and processed in an encrypted format, such as with cloud computing and storage. The cloud servers must embed some extra data directly in these encrypted pictures for the sake of content notation and/or tamper detection. Reversible data hiding in the encrypted domain is a developing technique that will be helpful in cloud computing since it can maintain anonymity. In this research, a brand-new data concealing strategy for encrypted pictures that is separable and error-free is suggested. After examining the interpolation technology's characteristics, sample pixels are encrypted using a stream cypher, and non-sample pixels' interpolation errors are encrypted using a specified encryption method. Then, the data-hider, who does not know the original image content, mayreversibly embed secret data into interpolation-error using a modified version of histogram shifting and difference expansion technique. In order to adapt to different application scenarios, data extraction can be done either in the encrypted domain or in the decrypted domain. In addition, real reversibility is realized, that is, data extraction and image recovery are free of any error. Experimental results demonstrate the feasibility and efficiency of the proposed scheme.

[10] **D. Xu et.al** proposed that Digital image sometimes needs to be stored and processed in an encrypted format to maintain security and privacy, e.g., cloud storage and cloud computing. For the purpose of content notation and/or tampering detection, the cloud servers need to embed some additional information directly in these encrypted images. As an emerging technology, reversible data hiding in the encrypted domain will be useful in cloud computing due to its ability to preserve the confidentiality. In this paper, a novel separable and error-free reversible data hiding scheme in encrypted images is proposed. After analyzing the property of interpolation technology, a stream cipher is utilized to encrypt sample pixels and a specific encryption mode is designed to encrypt interpolation-error of non-sample pixels. Then, the data-hider, who does not know the original image content, may reversibly embed secret data into interpolation-error using a modified version of histogram shifting and difference expansion technique. In order to adapt to different application scenarios, data extraction can be done either in the encrypted domain or in the decrypted domain. In addition, real reversibility is realized, that is, data extraction and image recovery are free of any error. Experimental results demonstrate the feasibility and efficiency of the proposed scheme.

[11] **K. Kainth et.al** As opposed to encrypting an image in its original pattern, this paper offers an alternative method in which the image is divided into entirely different elements and then combined into a pattern that is only known to authorised parties, followed by additional encryption and decryption processes. The method described in this study offers more security. In MATLAB, every encryption and decryption technique has been used, and an analysis has been made for a variety of quality of picture.

[12] **D. Xu et.al** suggested that in order to retain security and privacy, digital images may occasionally need to be saved and processed in an encrypted format, such as with cloud computing and storage. The cloud servers must embed some extra data directly in these encrypted pictures for the sake of content notation and/or tamper detection. Reversible data hiding in the encrypted domain is a developing technique that will be helpful in cloud computing since it can maintain anonymity. In this research, a brand-new data concealing strategy for encrypted pictures that is separable and error-free is suggested. After examining the interpolation technology's characteristics, sample pixels are encrypted using a stream cypher, and non-sample pixels' interpolation errors are encrypted using a specified encryption method. The data-hider can then employ an improved version of the histogram moving and difference expansion approach to reversibly encode sensitive data into interpolation error while hiding the original image's knowledge. Extraction of data can be performed either in the protected sector or in the uncovered domain to accommodate various application situations. Finally, true turnaround is achieved, meaning that both picture and data recovery are error-free. Results from demonstrations show that the suggested plan is feasible as well as successful.

[13] **K. Kainth et.al** Data becomes turned into a cypher text across the manner of confidentiality so that only authorised parties may access it. In basically, image encryption is an essential role of encryption where 2-D pictorial data is encrypted; as a result, the entire encryption process is carried out upon it. In this study, certain enhancements to an earlier method of picture encryption based on SCAN patterns are made in an easy-going way that drastically improves security. In the SCAN process, a special code called the four out of eight-code is used to create the airline image, which then gets transferred to the original picture to set up the encrypted image. Rather than encrypting an image in its original pattern, this paper provides another approach within which image is split into totally different elements so that it merged in to a pattern which is solely known to authorize parties and then further encryption and decryption process method is completed.

[14] **M Pooyan et.al** presented a novel method for digital audio steganography where encrypted covert data is embedded into the wavelet coefficients of host audio signal. To avoid extraction error we use lifting wavelet transform. For using the maximum capacity of audio signals, we calculate hearing threshold in wavelet domain. Then according to this threshold data bits are embedded in the least significant bits of lifting wavelet coefficients.
Inverse lifting wavelet transform is applied to modified coefficients to construct stego signal in time domain. Experimental results show that proposed method has large payload, high audio quality and full recovery.

[15] **S. Chakravarthy et.al** Now a days, the need for of data-integrity and privacy is higher than ever. From small bank transactions to large-scale classified military information, these factors have been prerequisites. This classified information can be faked deliberately by the sender with the help of a pseudo secret in order to increase the level of security. Cryptography makes this information unreadable and steganography hides the same. This paper brings together the paths of steganography and Rijndael encryption, which is commonly known as the AES. The secret image or text-file is encrypted using AES algorithm and is then camouflaged into the pseudo-secret image using a modified proposed Knight's Tour-inspired chess-based algorithm that enhances data diffusion. Then the pseudo-secret image/text is once again encrypted using the AES algorithm and is masked into the cover image/text using a checkerboard-based location map. Thus lossless, highly secure, nested-layer steganography is achieved with high complexity and high PSNR for various extension types of images as illustrated in the paper.

[16] **Rehana et al.** proposed LSB based video steganography which uses genetic algorithm visual cryptography for secure data hiding and transmission over networks. This approach uses the genetic algorithm to shuffle the pixel location of the stego image and visual cryptography to create the shares. Since visual cryptography is used, it suffers from the requirement to represent every pixel within the original frame by multiple pixels in every share, leading to a contrast deterioration problem called pixel expansion.

[17] Maggy Mashaley et all., there are hundreds of applications that need to secure their transmission, where a lot of techniques were implemented and applied for this reason. Attackers are always breaking these secure techniques to penetrate the communication channel. Steganography and cryptography are two approaches that provide the channel with data security. Steganography hides the data by inserting it into digital media e.g. images, videos, and audios. Cryptography encrypts the data in a way that makes it unreadable for anyone other than the receiver. In this work, a well–secured algorithm is implemented through integrating the two layers, the cryptography layer and the steganography layer. In cryptography we applied the 3– DES technique. In steganography we embedded the secret message in an image using an enhanced least significant bit (LSB) technique. We used the Peak Signal–to–Noise Ratio (PSNR), the mean squared error (MSE) and structure similarity index measure (SSIM) to evaluate the quality of the image after the embedding. In order to investigate the performance, we simulated our proposed algorithm on different images and the results showed that the PSNR reaches up to 75 dB and MSE of 0.0018. Moreover, the obtained results were compared with some of the popular LSB Steganographic methods.

[18] Krishnakanth Tiwari et all., method scatters the data to be infused into the image by randomly picking up the pixels and changing their LSB value accordingly. This random distribution makes it difficult for unknowns to look for the data. .In order to investigate the performance we simulated our proposed algorithm on different images Effective in all cases such as random embedding with LSB replacement and random embedding for sequential for LSB matching. In this paper, an information hiding method into codded images. Technique only applies to synthetic images with a small number of distinct colours such as logos and flags and it is not effective for big sizes images.

[19] Lalit, Lokesh et all., This paper proposed an android based secured system named Steg! developed by combining the cryptography and steganography. Here the algorithm used for cryptography is Advanced Encryption Standard (AES) and Least Significant Bit is used for the steganography. This hybrid approach increases the level of secretion of information from unauthorized the access by encrypting the message and hiding into the image. The application helps the user to hide/unhide the text to/from the image. The proposed system above is proven to be powerful and robust than those system which implements cryptography and steganography alone.

[20] Zimin jhang et all., to a large number of user-uploaded images, social networks have become secure channels for covert communication. However, the JPEG recompression of social networks changes the DCT coefficients of stego images, resulting in the failure of adaptive steganography. To achieve steganography in lossy channels, robust steganography has been proposed. In this letter, the ability against JPEG recompression of robust.

steganography is further improved by introducing a robustness cost function. For calculating the robustness cost, a robustness model based on the spatial domain calculated from DCT coefficients is firstly proposed.

[21] Amr A. Hanafy, Gouda I. Salama and Yahya Z. Mohasseb The paper likely discusses the technical details of their proposed model, including the algorithms used for encoding and decoding the hidden information, as well as any security measures employed to protect against detection and interception. It may also include experimental results demonstrating the effectiveness of their model. Overall, the paper appears to be focused on the development of a novel technique for secure communication, which may be of interest to researchers and practitioners in the fields of information security and cryptography.

[22] Punidha et al. used the concept of integer wavelet transform for sending audio speech signals through the steganography technique. The authors used the well-known Haar wavelet method with the integer wavelet transform for hiding secret messages. The authors used various objective parameters, such as signal-to-noise ratio, MSE, PSNR, and structural similarity index to evaluate the performance of the proposed method. The LL band of wavelet was used to store the data inside the image while the Daubechies method, along with the Haar, was used for comparison purposes. The algorithm with the Daubechies method performed better in comparison to the Haar wavelet method.

[23] Vardhan et al. proposed a reversible steganography technique that was based on the wavelet transform. They used the integer wavelet transform method on the encrypted digital images. The authors also performed mapping of integers with the cumulative density functions. Further, they used the sub-band of the encrypted image to store the secret message inside the cover image. The authors used the concept of histogram shifting for performing the steganography. The proposed method was compared with the other known steganography methods like logistic mapping and least significant methods. From the results, it was clear that the proposed method outperformed the other methods.

[24] **Yin et al.** performed steganography using reversible data hiding. The study performed steganography in encrypted digital images with the classification permutation. The permutation method used the XOR encryption and then data was embedded into the most significant bit of the encrypted image. With this approach, the visual quality of the recovered original image was very good. The authors compared the proposed algorithm with the other known methods of reversible data hiding like the Zhang method and Wu method.

With the results, it was shown that the proposed method showed lossless recovery of the original image even when the embedding rate was higher.

[25] **Manikandan et al.** used the concept of encryption for obtaining the image and message from the stego image using the reversible data hiding technique. The authors performed the research on medical digital images. They saved the data of patients into the concerned medical images of the same patient and the advantage of it was that there was no need of sending patient data into another file. The main aim of the proposed method was to get good data embedding capacity and the method should have a low bit error rate in comparison to other methods. In the encryption mechanism, the authors used three keys to share data between sender and receiver. From the results, it was shown that the proposed method had good embedding capacity and took less time of execution.

[26] **Anmol D Kulkarni** and his co-operative researcher presented an improved data security algorithm, to maintain quality of cover image and to reduce the size of video before transmission. In this paper, two stage processes is used to embed secret text data into a video clip. First stage is image steganography by using LSB method. Second stage is video steg steganography using DCT algorithm. The size of the video is increased after embedding process hence lossless compression technique is used. Advantages of this method are enhanced data security, visual quality of stego video remains unchanged and size of the final stego video is reduced for fast transmission.

[27] **R. Shanthakumari and Dr. S. Malliga** proposed video steganography using LSB Matching Revisited (LSBMR) algorithm. LSB Matching Revisited (LSBMR) algorithm selects the embedding regions according to the size of secret message and difference between two consecutive pixels in cover image. For lower embedding rates, only sharper edge regions are used while keep smoother regions as they are. In this method, LSB Matching Revisited algorithm is used to embed the secret message into the video. This method is analyzed in terms of both Peak Signal to Noise Ratio (PSNR) as well as Mean Square Error (MSE). Disadvantages of this method are lack of security and low embedding rate.

[28] **Ramadhan J. Mstafa and Khaled M. Elleithy** proposed a secure video steganography algorithm based on the principle of linear block code. In this method, nine uncompressed video sequences are used as cover data and binary image logo as a secret message. The pixel's position of both cover videos and a secret message are randomly recorded by using a private key to improve system's security. To improve security, the result of the encoded message will be Word with random generated values. Then the secret message is encoded by applying Hamming code (7, 4). Advantage of this method is, it is robust i.e. the embedding area in each frame is randomly selected and it will be different from other frames to improve the robustness. Security has been satisfied by having more than one key to embed and extract the secret message.

[29] **Dhande et al.** proposed a reversible steganography method that was based on an encryption mechanism. The main utilization of the proposed algorithm was for the gray scale digital images which could hide the images in the cover image with the help of a suitable encryption mechanism. The authors used two keys for performing steganography. One key was used for hiding the data and another for encrypting the data. The authors used the advanced encryption standard method for performing the encryption in the digital images while the least significant method of steganography was used to get the accuracy and efficiency in the proposed approach.

[30] **Elharrouss et al.** utilized k-least significant bit of cover image to perform steganography. Authors stored one image inside the other cover image. First, at the sending side, the most significant bit was selected which could be stored in the cover image, and in this way, a complete secret image was hidden in the cover image. On the decoding side, the authors used the concept of region detection. Here, an algorithm is used to search various region where the data of the secret image was hidden. Peak signal to noise ratio was used as an objective parameter. The proposed algorithm however performed below the expected level.

[32] Maninder Pal Singh, Harmandeep Singh have proposed Most Significant Bit(MSB). MSB, also called the high-order bit. The MSB is sometimes referred to the left-most bit due to the assembly in writing more important digits further to the left. The MSB can also equate to the sign bit of a binary number in one's or two's complement notation, "1" meaning negative and "0" meaning positive. In their technique the problem definition is they increase hiding capacity as compare to LSB. Firstly, a video file is read and Video is in any format like avi, mp4. The video consists of audio as well as images. On both parameter work can be done so according to requirement video information is extracted.

Chapter 3

ANALYSIS

3.1 Problem Statement

- In order to overcome the issues with existing approaches, we proposed the system "**EFFICIENT DATA HIDING IN VIDEO STEGANOGRAPHY USING LSB TECHNIQUE**".
- To create system which implement high security transmission of secret information using steganography for military information.
- In JPEG steganography detection of secret message was a hectic task.
- There are large number of information was not able to share in case of other encryption techniques
- Image steganography is not appreciated to use while large data needs to be shared secretly.

Video steganography using LSB (Least Significant Bit) algorithm in Java can encounter several problems, including:

Security: The LSB algorithm is a common and widely used method for video steganography, but it is vulnerable to attacks such as steganalysis. Attackers can detect the presence of hidden information by analyzing the video's LSB data, which may expose sensitive information.

Capacity: The amount of hidden data that can be embedded using the LSB algorithm depends on the video's size and complexity. In some cases, the amount of hidden data may be limited, which can reduce the effectiveness of the steganography technique.

Robustness: LSB steganography can be susceptible to distortion or noise, which may result in the loss of the hidden information. The technique must be robust enough to withstand various types of processing and transformations that video data may undergo.

Compatibility: The LSB algorithm may not be compatible with all video file formats and odes. The video file's characteristics may affect the algorithm's performance and may result in compatibility issues with different video players or devices.

Performance: The process of embedding and extracting data using the LSB algorithm can be computationally intensive, which may result in slower performance. In some cases, this may not be acceptable, especially for real-time video applications.

User Awareness: The use of video steganography using LSB algorithm requires some level of user awareness. Users must be aware of the presence of the hidden data, the location of the data, and the method of extraction. Failure to do so may result in the loss or exposure of sensitive information.

3.1 Objectives

- The primary objective of this method is to hide a lot of data with a high quality of stego video and accomplish a high security for the information hiding inside the cover video.
- The thought behind the LSB algorithm is to insert the bits of the hidden information into the least significant bits to the pixels.
- LSB algorithm is easy to implement and simple to understand.
- The LSB technique is quite robust and can withstand most of the common attacks and manipulations, such as compression, filtering, and noise addition.
- Video Steganography is used in covert communication, legal fields and copyright Control.
- Video steganography utilising Java's LSB methodology aims to offer a safe and reliable way to hide sensitive information within a video clip without compromising the video's integrity or quality.
- The LSB approach makes sure that the concealed data is embedded in the video file in a way that it cannot be seen by or heard by a human being and does not degrade the video file's quality.The LSB approach enables the concealment of a sizable quantity of data in a single video file without materially reducing the video's quality.
- Video steganography using LSB technique in Java is to provide a secure and robust method of concealing confidential information within a video file, without affecting the quality or integrity of the video.
- The LSB technique ensures that the hidden data is embedded in the video file in such a way that it is not visible or audible to the human eye or ear, and does not affect the quality of the video file.
- The LSB technique allows the hiding of a large amount of data in a single video file without significantly affecting the quality of the video.

3.2 Methodology

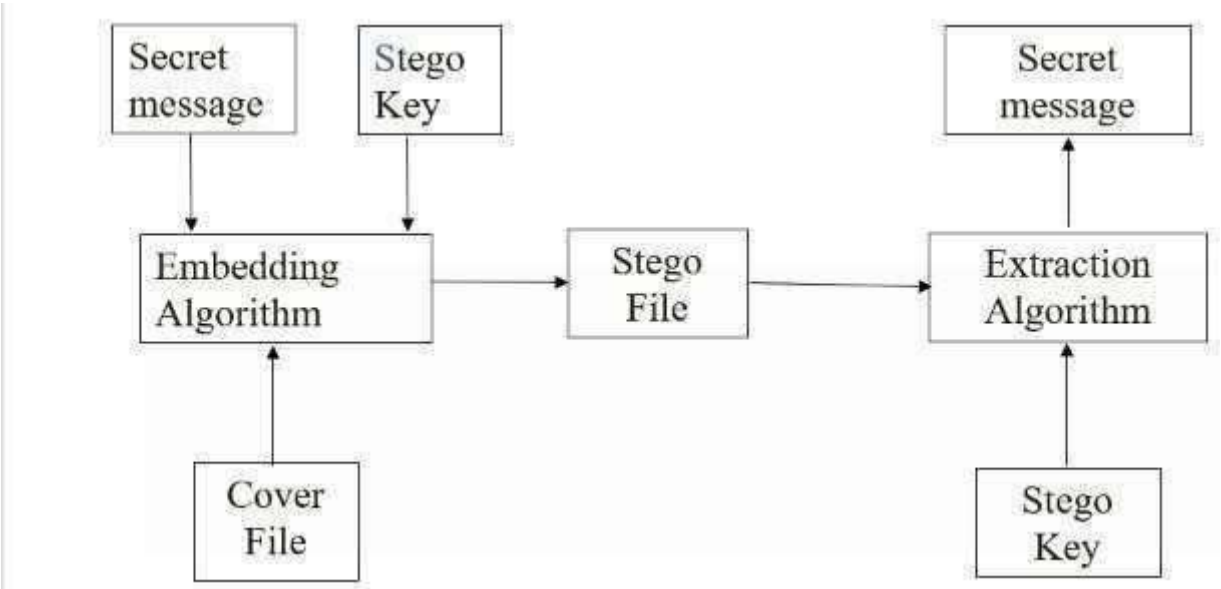


Fig 3.1: Methodology of video steganography

In this section, we proposed an algorithm which consists of random selection of frames, LSB method, and resizing an image file. The architecture is shown in the figure downwards. As previously, we also know video steganography can be done by audio or image steganography method, here we choose image steganography logic to implement video steganography. Here we take the video-path and the hiding image path as the input and also the key of the hidden technique method must be inputted. First of all, the video file is converted into frames. Generally, in small video .AVI, .MPEG, .MP4 video at least 20-25 frames can be generated per second. A frame in the video is actually an image consisting of a collection of pixel values (color and intensity) in a matrix form or in a list formation. The 24 - bitmap RGB image has 24 bits values for each pixel, 8-bits for each 3 color channels. The RGB is most suitable as there are lots of information where we can hide secret messages, with one bit change for each byte.

Bit-Slicing:

In this section, the information, needs to be hidden is taken for processing. This process includes image bit slicing and division of that image into 8 different bit-planes. The pixel values of each image are converted to its corresponding 8-bit binary values. Every ith bit is taken from each byte of pixel values to form the ith bit-plane image.

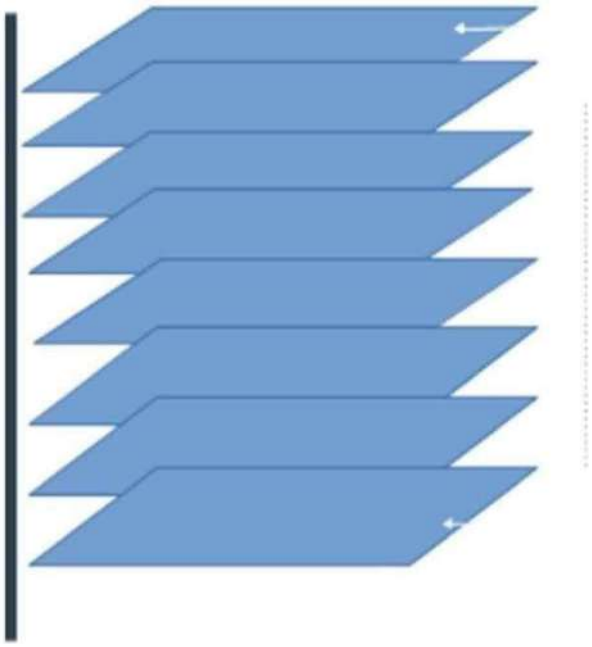


Fig 3.2: Bit slicing

The first bit-plane constructed from every first bit is called MSB plane while the last one is called LSB plane.

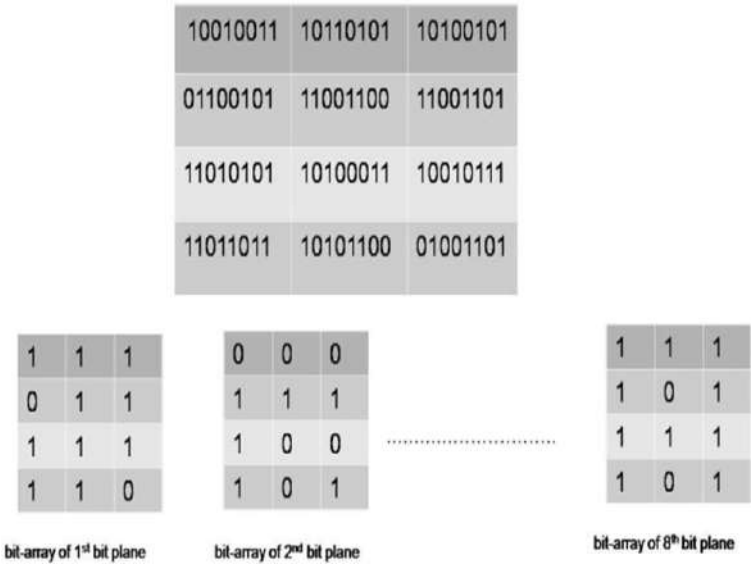
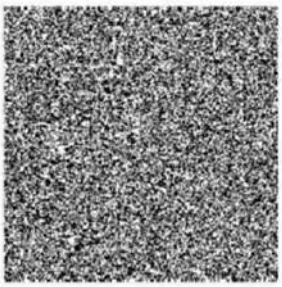


Fig 3.3: Bit plane

Original image(cameraman.tif)



MSB plane image

LSB image(8th bit image)

Fig 3.4:LSB and MSB image

LSB Technique

First of all, the video file is converted into frames. Generally, in small video .AVI, .MPEG,.MP4 video at least 20-25 frames can be generated per second. A frame in the video is actually an image consisting of a collection of pixel values (color and intensity) in a matrix form or in a list formation. The 24 - bitmap RGB image has 24 bits values for each pixel, 8-bits for each 3 color channels. The RGB is most suitable as there are lots of information where we can hide secret messages, with one bit change for each byte.

Each component is of one byte i.e., 8-bits in which the first one is the most significant bit. In LSB technique (Least Significant Bit) is used for hiding secret information resulting in the change in the last bit of each byte of the component. Substitution of the least significant bit results in human imperceptibility. We all know the first and foremost requirement of steganography is the invisibility of the changes in the cover frames. The strength of steganography lies in its ability to be unnoticed by the human eye. For hiding three bits of the data in every pixel's color, we used a 24-bit image. Human eye cannot easily differentiate between 21-bit color and 24-bit color.

Frame selection:

In the embedding process of this algorithm involves selecting frames randomly from the cover video in which the data is to be hidden. Here we used a random function which generates random numbers within a given range. We also used a value which is also the Key for retrieving the data, given by the user. By using this value selected frames can be selected and those frames are unique for that particular value. From those frames, users can retrieve the data i.e. bit-planes embedded in those frames. As we divided the original image into 8-bit planes, so 8 frames are sufficient to embed those bit-planes.

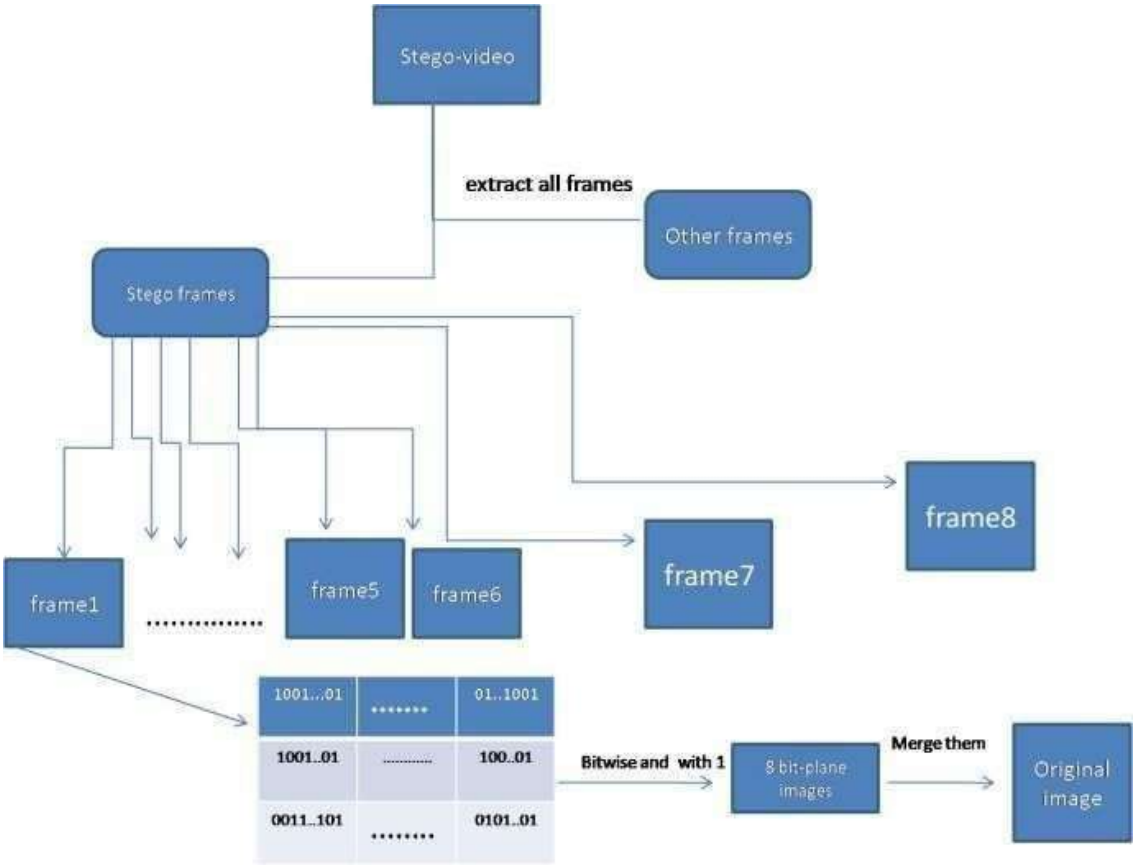


Fig 3.5: Process of Frame Selection

A. Steps of Image Embedding:

Step 1: Read the text and image file.

Step 2: Resize that image into the size of the cover video.

Step 3: Divided the images into bit-slicing method.

Step 4: Input the cover video.

Step 5: Split that video into frames.

Step 6: Select 8 frames where a bit sliced image will be embedded.

Step 7: Find that LSB bit of the cover frames.

Step 8: Embed those bit planes into that image frames and return those frames into its previous position.

Step 9: Regenerate video frames.

B. Steps of Image Extraction:

Step 1: Input the Stego video.

Step 2: Extract frames from the video.

Step 3: Find those frames by the specified key

Step 4: Find LSB of those frames.

Step 5: Extract those bits from those stego frames.

Step 6: Merge those bits to form the final image.

Step 7: Extract the Dual Encrypted Text from Image.

3.3 Software Requirement Specification

a. Operating system: Windows XP Professional

b. Front End : JAVA.

c. Tool : Eclipse 3.3

3.4 Hardware Requirement Specification

- System : Pentium IV 2.4 GHz.
- Hard Disk : 40 GB.
- Floppy Drive : 1.44 Mb.
- Monitor : 15 VGA Color.
- Mouse : Logitech.
- Ram : 256 Mb.

Chapter 4

SYSTEM DESIGN

The design of systems implies the technique of developing a system's architecture, modules, interfaces, and data to meet specific demands. The application of systems theory to product development is what is known as systems architecture.

Systematic system design is implied by the term systems design. In either case, the process is systematic and takes into account all relevant system variables, involving the architecture, vital hardware, and software, right down to the data and how it moves by way of the system's alterations. The approach may be from below or from the top, but it will always include careful evaluation of all associated variables. Then, there is overlap between systems design, engineering systems, and networks architecture.

4.1 System Architecture

4.1.1 Class Diagram

In software engineering, a class diagram in the Unified Modelling Language (UML) is a kind of static structural diagram that explains the structure of a system by displaying the system's classes, their properties, operations, and connections among objects. In object-oriented modelling, the class diagram serves as the primary building component. When turning the models into computer code, it is utilised for both overall conceptual modelling of the application's structure and for specific modelling. The data planning process can also employ class diagrams. A class diagram uses classes to depict both the classes that will be coded and the application's key components, interactions, and components.

In the diagram, classes are represented with boxes that contain three compartments:

- The top compartment contains the name of the class. It is printed in bold and centered, and the first letter is capitalized.
- The middle compartment contains the attributes of the class. They are left- aligned and the first Letter Lowercase.

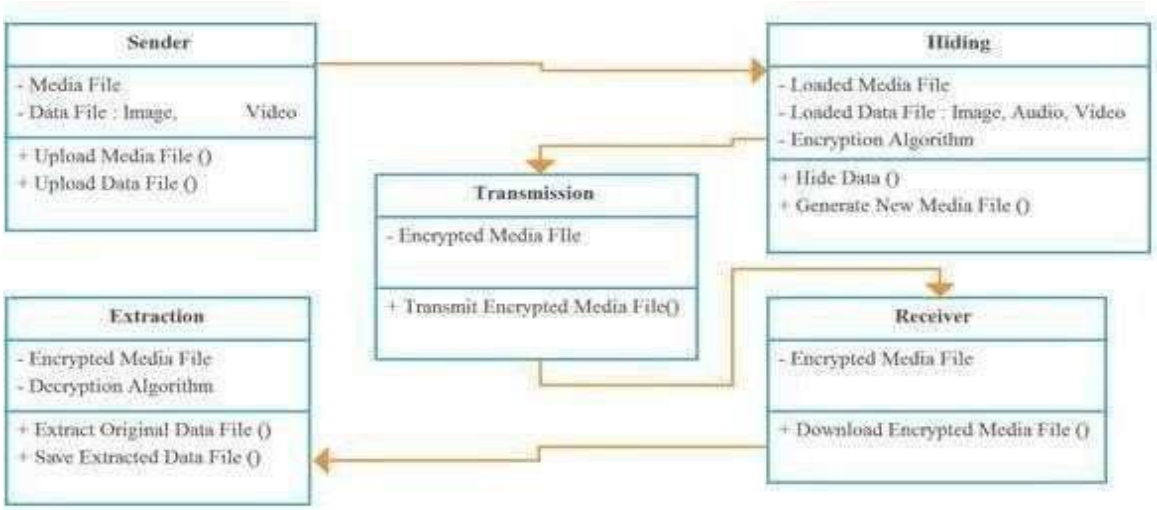


Fig 4.1: Class Diagram

4.1.2 Data Flow Diagram

A dataflow outline is a tool for referring to knowledge progression from one module to the next module as shown. This graph gives the data of each module's info. The map has no power flow and there are no circles at the same time.

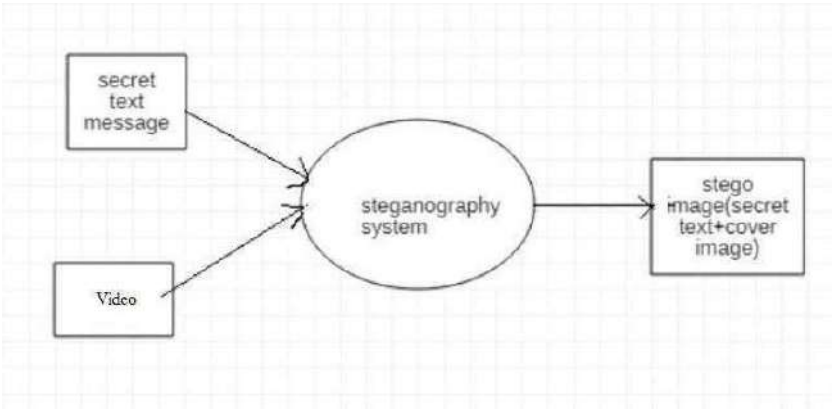


Fig 4.2: Data flow level 1

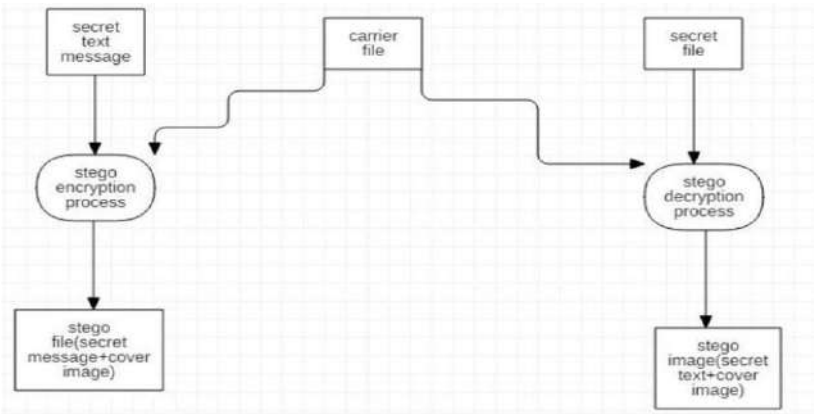


Fig 4.3: Data flow level 2

4.1.3 Use Case Diagram

A Use Case Diagram is a lot of situations that reflect a client-frame relationship. A use case chart shows the entertainer-to-use relationship. Usage cases and on-screen characters are the two main elements of an usage case diagram. An on-screen character refers to an user or other person connected with the demonstrated process. A use case chart in figure 4.3.3 is an out-of - the-box perspective that speaks to some activity each module will perform to complete an errand. A use case is a methodology used in system analysis to identify, clarify and organize system requirements.

In this context, the term "system" refers to something being developed or operated, such as a mail-order product sales and service Web site. Use case diagrams are employed in UML (Unified Modeling Language), a standard notation for the modeling of real-world objects and systems. There are a number of benefits with having a use case diagram over similar diagrams such as flowcharts.

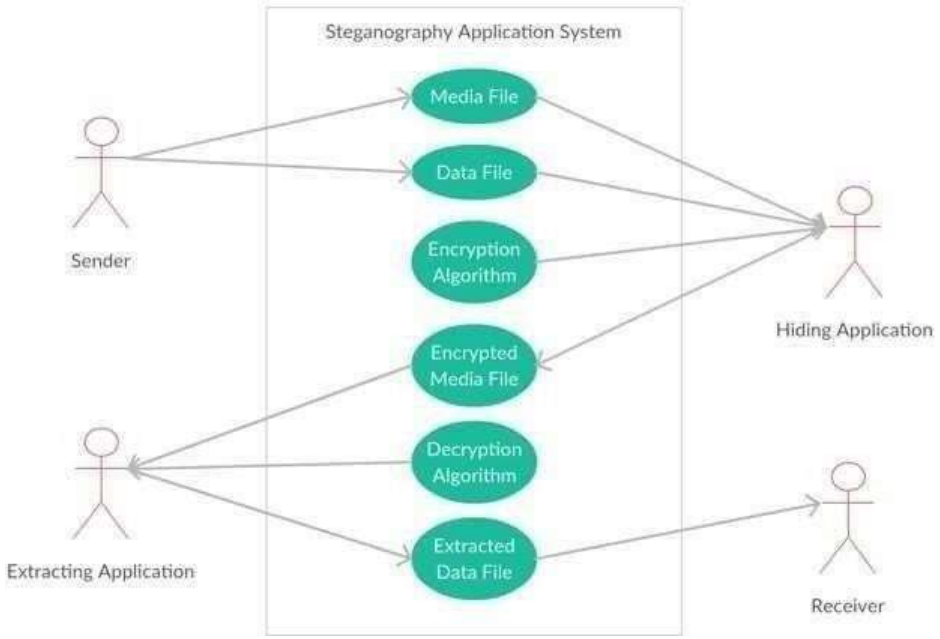


Fig 4.4: Use case Diagram

4.1.4 Sequence Diagram

A sequence diagram is a type of interaction diagram because it describes how—and in what order—a group of objects works together. These diagrams are used by software developers and business professionals to understand requirements for a new system or to document an existing process. Sequence diagrams are sometimes known as event diagrams

or event scenarios. A usage scenario is a diagram of how your system could potentially be used.

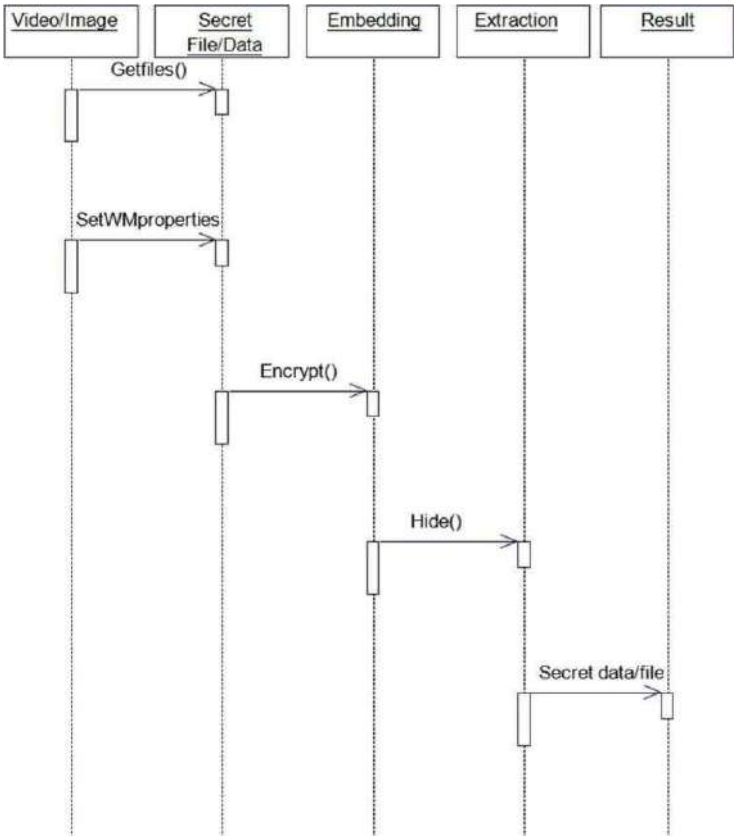


Fig 4.5: - Sequence Diagram

4.1.5 Activity Diagram

The fundamental goals of activity diagrams are the same irrespective of the other four types. It performs perform well at reflecting the system's dynamic nature. An activity diagram is used to show how signals go from one activity to the next. The other four styles are utilised to show the message flow from one item to the next. A specific type of system operation is called an activity. Activity diagrams may be used to both show how dynamic a system is and to create a true executable system using both forward and reverse engineering strategies. The message section is the only portion of the activity diagram that is missing. There is no proof that via a communication channel, one action conduces to another. The last one goes by the name of a flowchart.

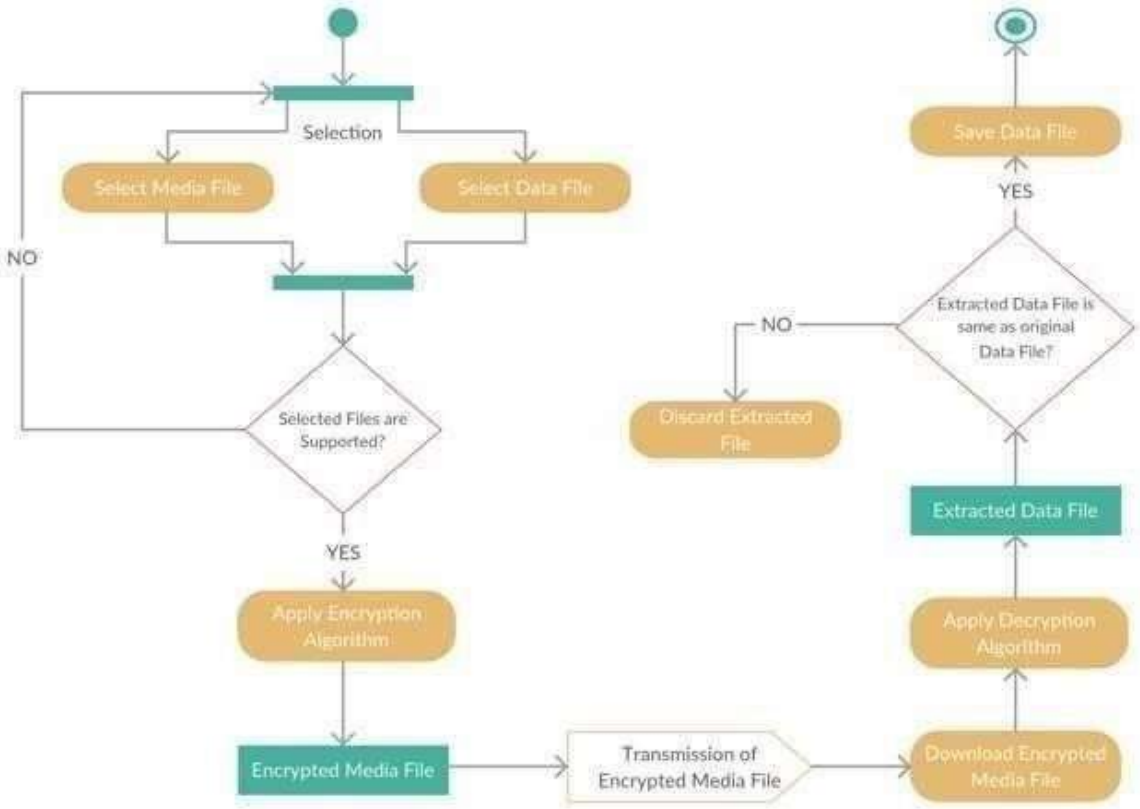


Fig 4.6: - Activity Diagram

4.2 MODULES USED

The video stego machine performs the process of conceal and reveal in following modules. modules of Video Stego machine are

- Video Header Information
- File Handling
- Encryption
- Steganography – Conceal data
- DeSteganography - Reveal original data
- Decryption
- Graphical User Interface

A. Video Header Information

The Resource Interchange File Format (RIFF) document format serves as the basis for verifying the AVI (Audio/Visual Interleaved) format of a carrier file. It contains essential header information collected by the video header module, including details about the AVI Main Header, AVI Stream Header, Audio, and BITMAP. This module stores information that helps determine if the carrier file is in the AVI format and identifies whether it is a video, audio, or another type of format.

B. File Handling

The AVI (Audio/visual interleaved) file header is not opened during file handling, and the file's contents are processed in ASCII format. This provides a key file by the reading the AVI file in terms of bytes corresponding to the header. The intended text file for embedding will be transformed into a binary value. By placing zeros in front of each bit, the binary value is subsequently transformed to a similar 8-bit value.

C. Encryption

To deceive curious individuals the message that will be concealed inside the carrier file is encrypted with a key. By doing this, the security of sharing data can be increased. The Stego machine is strong because of this powerful encryption method. The input message is initially shifted to a byte value in this module. The user provides the key, which is appended to the relevant byte and saved in a separate byte array prior it's converted to character to construct the message's encrypted form. The plain text message and a key value to encrypt the message are provided to this function.

D. Steganography – Conceal data

This module carries out the steganography procedure. Here, the length of the carrier file (an AVI file) is collected and verified to see if it is eight times longer than the text file's length. Create a key file by writing the content of the AVI file from the data starting point to the finish. Find the starting point of the data in the AVI file. The carrier file is binary-encoded. The outcome is written into the freshly produced text file as well as rewritten to the data portion of the AVI file. A stego'd video clip plus a key file that must be transferred via a secure channel make up the output of this system. Fig. 2 shows a clear illustration of hiding the data.

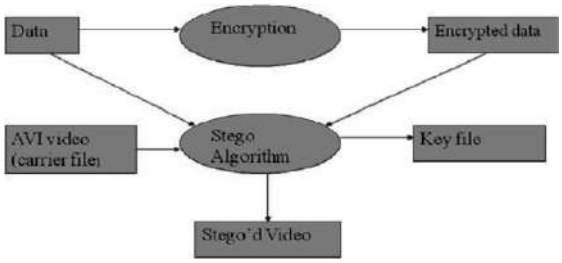


Fig4.7: Steganography of Stegomachine

E. DE Steganography- Reveal Original data

The video file is decoded by this DE Steganography module, which then extracts the video's concealed data. Here, the inputs are the carrier file (an AVI file) and the key file. To determine the beginning of the data in the AVI file, the Key file and the AVI file are opened in random access mode. This compares the differences between the Key file and the AVI file by reading each byte sequentially. An original AVI video file and a data file with the message that was concealed inside the AVI video file are the outputs that were retrieved. The process to follow for tearing off the original data from a Stego'd video clip is highlighted in Fig. 3.

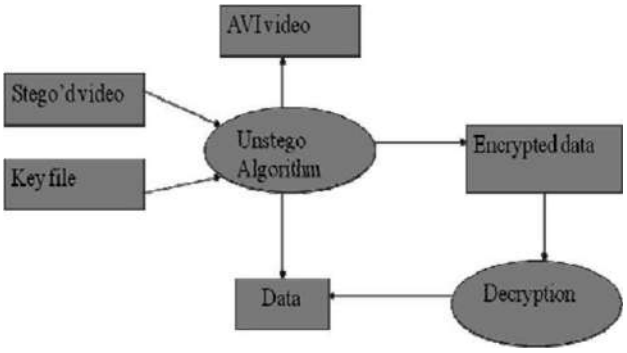


Fig4.8: DeSteganography of Stegomachine

F. Decryption

The key is used to decode the concealed message since, once the technique is known, it can be used to decrypt any encrypted data. The input message is initially converted to a byte value by this module. The user provides the key, which is deducted from each byte and saved in its own byte array before converted to characters to produce the message's decrypted form. The password protected message file and a key for decryption value are the inputs for this function.

G. Graphical User Interface (GUI)

This GUI serves as an intuitive wizard which can be employed by those without previous technical expertise. It allows an individual to execute both key management and steganography without using protection. This will aid the user in obscuring an expression in a video file via a wizard. Retrieve the hidden message in a stego’d video.

- Encrypt a text file.
- Decrypt an encrypted file.

Chapter 5

IMPLEMENTATION

The project's implementation phase is when the design idea is transformed into a functional system. As a result, it may be said to be the stage that will determine if a new system is a success and whether its users have faith in its ability to function well. The implementation step entails meticulous planning, research into the current system and its implementation limitations, creation of transitioning procedures, and an examination of transitional methods.

5.1 Overview of System Implementation

The implementation overview of video steganography using LSB encrypted by DES, Triple DES, and MD5 algorithm involves the following steps:

- Data Encoding: The data to be hidden is first encrypted using DES, Triple DES, or MD5 algorithm. This ensures that the hidden data is secure and can only be accessed by those who have the appropriate decryption key.
- LSB Embedding: The next step is to embed the encrypted data in the video file using the LSB technique. In this technique, the least significant bit of each pixel in the video file is modified to contain a bit of the hidden data. This process is repeated for several pixels in the video file, depending on the amount of data that needs to be hidden.
- Decryption and Extraction: To retrieve the hidden data, the video file is first decrypted using the appropriate decryption key for the encryption algorithm used. Once the video file is decrypted, the LSB technique is used to extract the hidden data by reading the least significant bits of the pixels. This data is then decrypted using the same decryption key used in the first step to retrieve the original data.
- Integrity Checking: Finally, an integrity check is performed on the extracted data using the MD5 algorithm to ensure that the data has not been tampered with or modified during transmission or storage.

Overall, this implementation ensures that the hidden data is both secure and imperceptible, while also allowing for efficient and reliable extraction of the data when required.

5.2 Algorithm

Data Encryption Standard (DES):

This module consists of same as Encryption and Decryption part using DES algorithm. The Data Encryption Standard (DES) is a block cipher that uses shared secret encryption. The DES algorithm uses a key of 56-bit size. Using this key, the DES takes a block of 64-bit plain text as input and generates a block of 64-bit cipher text. The DES process has several steps involved in it, where each step is called a round. Depending upon the size of the key being used, the number of rounds varies. For example, a 128-bit key requires 10 rounds, a 192-bit key requires 12 rounds, and so on. The algorithm process breaks down into the following steps:

- The process begins with the 64-bit plain text block getting handed over to an initial permutation (IP) function.
- The initial permutation (IP) is then performed on the plain text.
- Next, the initial permutation (IP) creates two halves of the permuted block, referred to as Left Plain Text (LPT) and Right Plain Text (RPT).
- Each LPT and RPT goes through 16 rounds of the encryption process. Finally, the LPT and RPT are rejoined, and a Final Permutation (FP) is performed on the newly combined block. The result of this process produces the desired 64-bit ciphertext.

Triple DES:

This module uses the Triple DES algorithm for its encryption and decryption aspects. The Data Encryption Standard (DES) cypher algorithm is employed three times to each data block using the Triple Data Encryption Algorithm (TDEA or Triple DEA), additionally referred to as Triple DES. The user must first generate and share a 3TDES key K, which is made up of the three separate DES keys K1, K2, and K3, before that use 3TDES. The real 3TDES key therefore has a length of 356, or 168 bits.

The encryption-decryption process is as follows:

- Encrypt the plaintext blocks using single DES with key K1.
- Now decrypt the output of step 1 using single DES with key K2.
- Finally, encrypt the output of step 2 using single DES with key K3.
- The output of step 3 is the ciphertext.

- Decryption of a ciphertext is a reverse process. ➢ User first decrypt using K3, then encrypt with K2, and finally decrypt with K1.

5.3 Pseudocode

Pseudo code is a term which is often used in programming and algorithm-based fields. It is a methodology that allows the programmer to represent the implementation of an algorithm. Simply, we can say that it's the cooked-up representation of an algorithm. Often at times, algorithms are represented with the help of pseudo codes as they can be interpreted by programmers no matter what their programming background or knowledge is. Pseudo code, as the name suggests, is a false code or a representation of code which can be understood by even a layman with some school level programming knowledge.

```
// Main Method
public static void main(String[] args) {


    StegoStick inst = new StegoStick();
    inst.setVisible(true);
}

public StegoStick() {
    super("Video Stegnography Using LSB");

    setLocation(300,300);
    setResizable(false);
    initGUI();
```

Fig 5.3.1 Implementation of main function

```
private void initGUI() {
    try {
        setDefaultCloseOperation(WindowConstants.DISPOSE_ON_CLOSE);
        getContentPane().setLayout(null);
        {
            titleLabel = new JLabel();
            getContentPane().add(titleLabel);
            titleLabel.setText("Video Stegnography Using LSB");
            titleLabel.setFont(new java.awt.Font("Baskerville Old Face",1,20));
            titleLabel.setHorizontalAlignment(SwingConstants.CENTER);
            titleLabel.setBounds(0, 0, 595, 49);
        }
        {
            containerPane = new JTabbedPane();
            getContentPane().add(containerPane);
            containerPane.setBounds(0, 49, 595, 371);
            containerPane.setTabPlacement(JTabbedPane.LEFT);
            containerPane.setFont(new java.awt.Font("Corbel",0,16));

            {
                hidingPanel = new HideGUI();
                containerPane.addTab("Hiding", null, hidingPanel, null);
                hidingPanel.setPreferredSize(new java.awt.Dimension(459,371));
                hidingPanel.setOpaque(false);

                unhidingPanel = new UnhideGUI();
                containerPane.addTab("Unhiding", null, unhidingPanel, null);
                unhidingPanel.setOpaque(false);
            }

        }
        pack();
        this.setSize(610, 461);
    } catch (Exception e) {
        e.printStackTrace();
    }
}
```

Fig 5.3.2 Implementation of GUI

Fig 5.3.1 and 5.3.2 represents the implementation of main function and GUI, in which we are setting the visibility true so that the window is always visible and calling the init GUI function.

```
public void setAlgorithm(int choice){
    encryptTechnique = choice;

    switch(choice){
        case 1: algorithm = new String("PBEWithMD5AndDES"); //DES
                break;
        case 2: algorithm = new String("DESede/CBC/PKCS5Padding"); //Triple DES
                break;
        default: algorithm = new String("PBEWithMD5AndDES"); //DEfault
                break;
    }
}
```

Fig 5.3.3 Algorithm Selection

```
private void InitiateCipher(int mode){
    try{
        if(encryptTechnique!=2){    // if not triple des
            paramSpec = new PBEParameterSpec(salt, count);
            keySpec = new PBEKeySpec(password.toCharArray());
            secretKey = SecretKeyFactory.getInstance(algorithm).generateSecret(keySpec);
        }
```

Fig 5.3.4 DES Algorithm

Fig 5.3.4 represents the selection of algorithms using the switch cases, so that if we choose 1 that is DES algorithm, in the same way, 2 means TRIPLE DES and default is DES and Md5. Fig 5.3.4 represents the implementation of DES algorithm by using only one key.

```
else{
    key = toKey(password);
    // Create a DESede key spec from the key
    DESedeKeySpec spec = new DESedeKeySpec( key );
    // Get the secret key factor for generating DESede keys
    SecretKeyFactory keyFactory = SecretKeyFactory.getInstance("DESede");
    secretKey = keyFactory.generateSecret(spec);
}
```

Fig 5.3.5 Triple DES Algorithm

Fig 5.3.5 represents the implementation of Triple DES algorithm by using a key, where key is generated by appending the same key, until the length is less than 25. Triple DES is the

common name for the Triple Data Encryption Algorithm (TDEA or Triple DEA) block cipher, which applies the Data Encryption Standard (DES) cipher algorithm three times to each data block.

```
    public byte[] toKey(String password){
        int pwdlen = password.length();

        int i=pwdlen;
        while(i < 25){
            password = password + password;
            i = password.length();
        }
        byte[] encryptKey = password.getBytes();

        return encryptKey;
    }
}
```

Fig 5.3.6 Password generation for Triple DES

Fig 5.3.6 represents the password generation for Triple DES, where it takes the password from the user of certain length, if it is less than 25 means, the same password is appended until its length becomes 25.

```
    fileChooser.setFileFilter(imgFilter);
    fileChooser.setDialogTitle(new String("Select Cover Image File"));
    int result = fileChooser.showOpenDialog(PasswordHidePanel.this);

    if (result == JFileChooser.APPROVE_OPTION) {
        File f = fileChooser.getSelectedFile();
        currentPath = f.getPath();
        coverField.setText(f.getAbsolutePath());
    }
}
});
```

Fig 5.3.7 Cover file selection

Fig 5.3.7 represents the selection of cover file using file chooser module which allows us to choose the image or video.

```
public class ConvertToBMP{
    public boolean convertToBMP(String imgFile){
        try{
            BufferedImage img = ImageIO.read(new File(imgFile));

            ImageIO.write(img,new String("bmp"),new File("C:\\WINDOWS\\Temp\\temp.bmp"));
            return true;
        }catch(Exception ex){
            return false;
        }
    }
    void deleteTempImage(){
        try{
            File f = new File("C:\\WINDOWS\\Temp\\temp.bmp");
            f.delete();
        }catch(Exception ex){
        }
    }
}
```

Fig 5.3.8 Conversion to .BMP format

Fig 5.3.8 represents the conversion of all the image formats such as .jpeg, .png, .jpg, .gif to .BMP (bitmap) format, as it is supported by the algorithms.

```
    int encrypt(String inputFileName, String password){
        this.password = password;
        try{
        //      JOptionPane.showMessageDialog(null,"algorithm: "+algorithm);
            InitiateCipher(Cipher.ENCRYPT_MODE);
            FileInputStream fin = new FileInputStream(inputFileName);
            BufferedInputStream in = new BufferedInputStream(fin);

            outputFileName = new String("C:\\Windows\\Temp\\tempe") + getExtension(inputFileName);
     //     String outputFileName = new String("tempe") + getExtension(inputFileName);

            FileOutputStream fout = new FileOutputStream(outputFileName);
            CipherOutputStream out = new CipherOutputStream(fout, cipher);
            int buffer;
            while ((buffer = in.read()) != -1)
                                    out.write(buffer);
            out.close();
            in.close();

            return 0;
        }
        catch(Exception ex){
            return 1;
        }
    }
```

Fig 5.3.9 Encryption

Fig 5.3.9 represents the implementation of encryption, where it reads the text and file by fileInputStream and BufferInputStream.

```
int decrypt(String inputFileName, String dstPathName, String password){
    this.password = password;

    try{
        InitiateCipher(Cipher.DECRYPT_MODE);
        FileInputStream fin = new FileInputStream(inputFileName);
        CipherInputStream in = new CipherInputStream(fin, cipher);

        outputFileName = new String(dstPathName + "\\tempd") + getExtension(inputFileName);
        //      String outputFileName = new String("tempd") + getExtension(inputFileName);
        FileOutputStream fout = new FileOutputStream(outputFileName);
        BufferedOutputStream out = new BufferedOutputStream(fout);

        int buffer;
        while ((buffer = in.read()) != -1)
                                out.write(buffer);
        in.close();
        out.close();

        return 0;
    }catch(Exception ex){
        return 1;
    }
}
```

Fig 5.3.10 Decryption

Fig 5.3.10 represents the implementation of decryption, where it reads out the text and file by fileInputStream and BufferInputStream.

5.4 Implementation Support

There are many features in Java, some of which are discussed below as follows:

1. **Simple:** Java is a simple programming language and easy to understand because it does not contain complexities that exist in prior programming languages the concept of pointers has been completely removed from Java which leads to confusion for a programmer and pointers are also vulnerable to security.

2. **Object-oriented:** Java is an Object-Oriented Programming Language, which means in Java everything is written in terms of classes and objects. The object is nothing but a real-world entity that can represent any person, place, or thing and can be distinguished from

others. Every object near us has some state and behavior associated with it.

3. **Platform Independent**: The design objective of java soft people is to develop a language that must work on any platform. Here platform means a type of operating system and hardware technology. Java allows programmers to write their program on any machine with any configuration and to execute it on any other machine having different configurations.

4. **Portable:** The WORA (Write Once Run Anywhere) concept and platform-independent feature make Java portable. Now using the Java programming language, developers can yield the same result on any machine, by writing code only once. The reason behind this is JVM and bytecode. Suppose you wrote any code in Java, then that code is first converted to equivalent bytecode which is only readable by JVM.

5. **Robust:** The Java Programming language is robust, which means it is capable of handling unexpected termination of a program. There are 2 reasons behind this, first, it has a most important and helpful feature called Exception Handling. If an exception occurs in java code, then no harm will happen whereas, in other low-level languages, the program will crash.

Basic syntax of Java

Java syntax can be executed by writing directly in the Command Line:

```
public class MyFirstJavaProgram { public static
        void main(String []args) {
        System.out.println("Hello World"); }
        }
C:\> javac MyFirstJavaProgram.java
```

```
C:\> java MyFirstJavaProgram Hello
World
```

Java Variables

In Java, variables are created when you assign a value to it. Java programming language requires variables to operate and handle data. Java creates several variables as per data format and data types. The variable declaration means creating a variable in a program for operating different information.

Example: int x = 5

String s = “Hello, World!”

Java Comment

Java has commenting capability for the purpose of in-code documentation. Single-line comments start with // and ends with //, and Multi-line command starts with /* and ends with */

Example:

#print (“Hello, World!”)

Chapter 6

TESTING

Testing can be stated as the process of verifying and validating whether a software or application is bug-free, meets the technical requirements as guided by its design and development, and meets the user requirements effectively and efficiently by handling all the exceptional and boundary cases.

The process of software testing aims not only at finding faults in the existing software but also at finding measures to improve the software in terms of efficiency, accuracy, and usability. It mainly aims at measuring the specification, functionality, and performance of a software program or application.

A trial run is a set of gestures used to verify a certain feature or competency of your computer system. For the purpose of validating any requirement, a test case is made up of test methods, test information, prerequisites and postconditions. A testing engineer may examine anticipated and actual findings and determine if a software product complies with the requirements using the specific variables or conditions of the test case.

6.1 Unit Testing

Designing test cases for unit testing ensures that the core programme logic is working correctly and that programme inputs result in legitimate outputs. It is important to verify the internal code flow and all decision branches. It is the testing of the application's separate software components. Before integration, it is done following the completion of each individual unit. This is an intrusive structural test that depends on understanding how it was built. Unit tests carry out fundamental tests at the component level and examine a particular configuration of a system, application, or business process. Unit tests make assurance that each distinct route of a business process adheres precisely to the stated specifications and has inputs and outputs that are well-defined.

Unit testing is often done as part of a combined code and unit test phase of the software lifecycle, while it is not uncommon for the two activities to be carried out as distinct phases.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

- **Features to be tested**
- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

6.2 Integration Testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problem

Integration Testing Software is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects. The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully.

6.3 System Testing

System testing guarantees ensures an integrated software system as a whole operates in harmony with requirements. It assesses a set up to be able to deliver known and predictable consequences. A case study of testing systems is the configuration-oriented system connection check. System testing is based on process flows and descriptions, with an emphasis on junctions and links that have already been pre-driven. System testing is a type of software evaluation that examines an entire whole to see how well it satisfies the required standards.

Integration tests successful modules are used as input after system testing integration testings aim is to find any discrepancies between the units that are combined system testing finds flaws in the integrated modules as well as the entire system a component or systems behaviour seen during verification is a result of system assessment system testing is done on the entire system under the instruction of operational or technical specifications or under the guidance of both the design behaviour and buyers expectations of the system are all checked during execution beyond the parameters specified in the programme specification srs it is used to test the system. In essence, system testing takes place out by an evaluation group that is separate from the programming team and helps to independently evaluate the state of the system. It was thoroughly tested in both practical and ineffective ways.

White Box Testing

White box testing is an instance of assessing where the application is tested. The tester is familiar with the inside workings, framework, and syntax of the software, or at the at least, knows precisely what it is intended to do. It has a goal. It makes use of to test regions that are inaccessible from a black box perspective.

Black Box Testing

Testing software in a black box is doing so without have a grasp of the inner workings architecture or language of the module being tested black box tests like the majority of other sorts of tests must be constructed from an obvious origin document such as a specification or criteria paperwork it is a type of testing where the element of technology being tested is handled like a mystery it is impossible to see inside without taking into account how the computer programme functions the test develops sources and replies to outputs.

6.4 Acceptance Testing

User Acceptance by users Any project's evaluation phase is crucial and necessitates the end user's active involvement. Additionally, it makes that the system satisfies the functional specifications.

Results of the tests: All of the above-mentioned test scenarios were successful. There were no faults discovered.

6.5 Functional testing

Functional tests offer methodical proof that the characteristics being tested are available in accordance with the physical and commercial needs, system documentation, and user manuals. Focus of functional testing is on the following areas:

- Acceptable Valid Input: Recognised Classes of Valid Input.
- Identifying classifications of incorrect input must result in their rejection.
- Functions: The assigned functions need to be used.
- Application outputs must be tested according to the classes that have been identified.

Systems/Procedures: It is essential to call interacting systems or processes. Functional tests are organised and prepared with a focus on requirements, important efficiency or unique test cases. likewise testing must take into account systematic coverage of data fields, established guidelines, and afterwards procedures besides business process flows.

Chapter 7

DISCUSSION OF RESULTS

This section discusses the results from each module implemented.

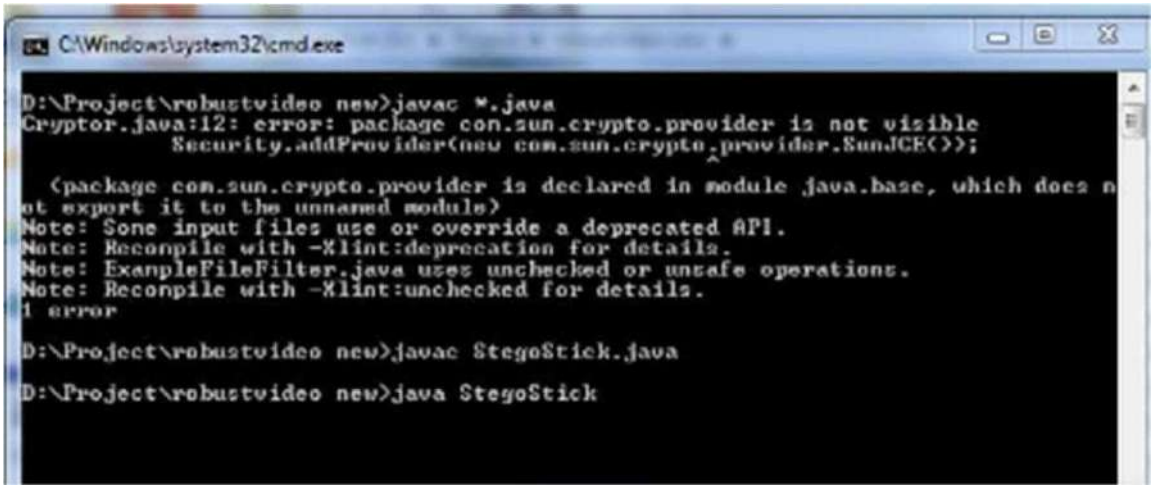


Fig 7.1 Compilation of StegoStick java file in command prompt

Fig 7.1 represents the compilation and execution of the Steg Stick file which is the main java file for the program execution in the command prompt. Type java StegoStick.java' for compilation. If there are no errors in, the command prompt will take to the next line. Now, type ' java StegoStck ' to run program.

Fig 7.2 Home Page

Fig 7.2 represents the home page, which gives a detailed description of all the algorithms used and enables to choose the algorithm. It also enables us choose whether to hide and unhide the secret message n the form of text or image

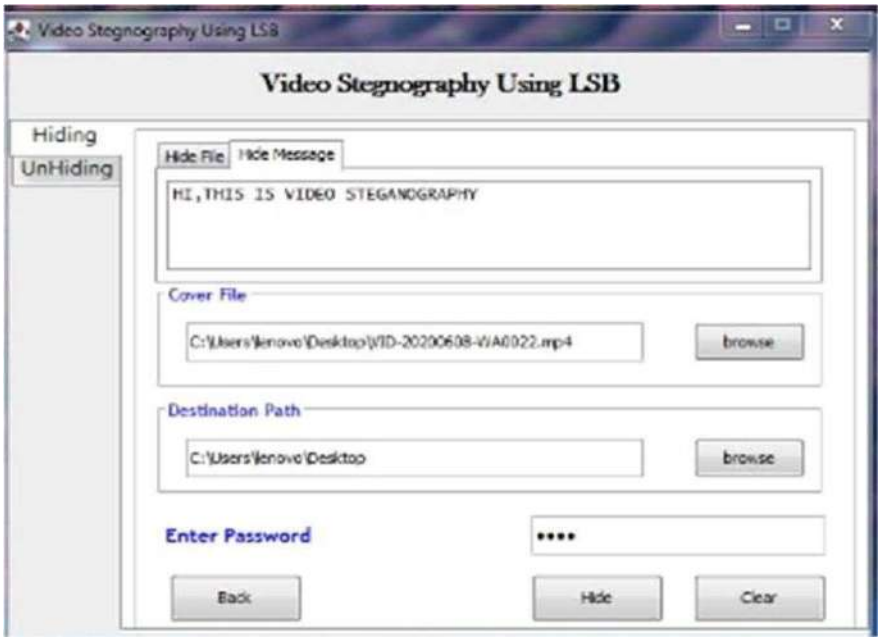


Fig 7.3 Hiding of Secret message in the form of text

Fig 7.3 represents the page of hiding the secret message in the form of text by giving the input, such as the secret message as text to be hidden and then the cover file as video to embed the secret message and the destination path for the output to be stored and finally the key.

Fig 7.4 Generation of steg file

Fig 7.4 represents the generation of steg file, in which the secret message is embedded into the cover file using LSB technique.

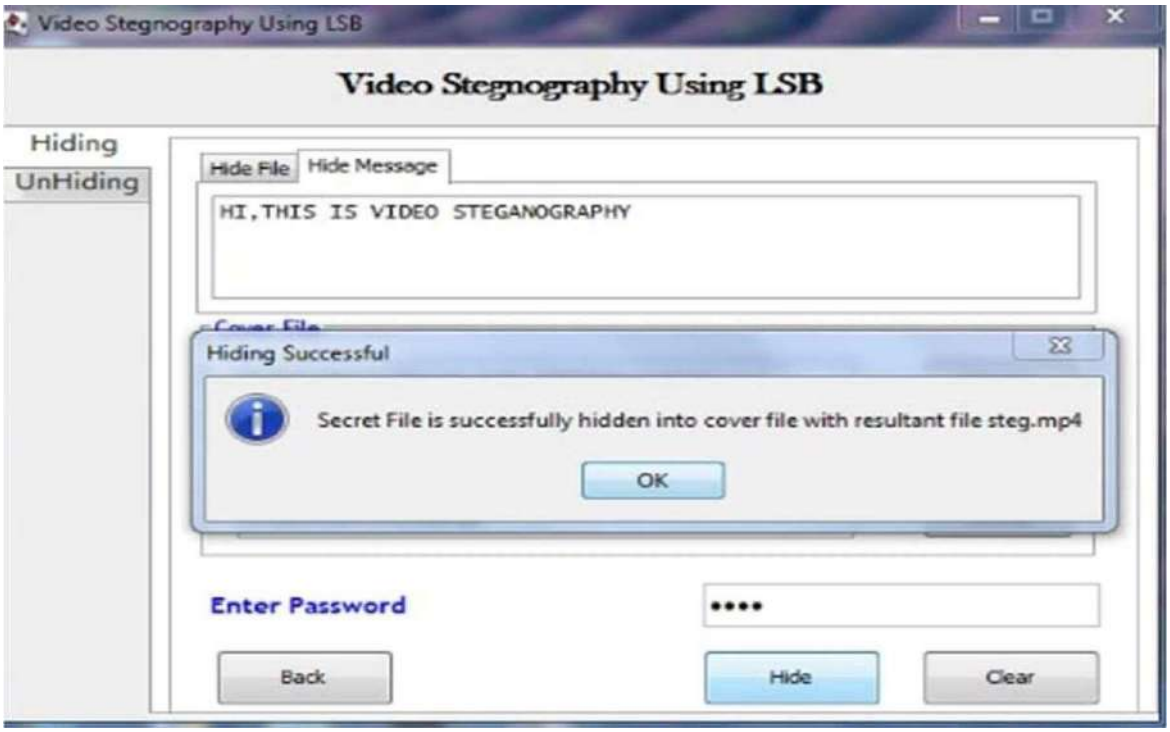


Fig 7.5 Pop-up message of successful hiding

Fig 7.5 represents the pop-up message of successfully hiding the secret message into the cover file.

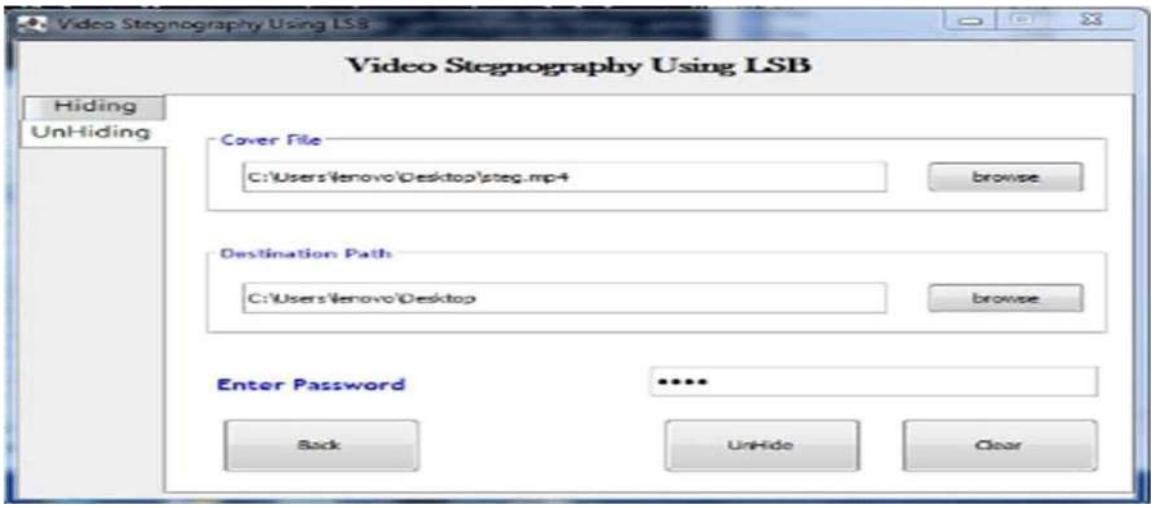


Fig 7.6 Unhiding of Secret message in the form of text

Fig 7.6 represents the page of unhiding the secret message in the form of text by giving the input, such as steg file which is the combination of both the secret message and the cover file, the cover file as video and the destination path for the output to be stored and finally the key.

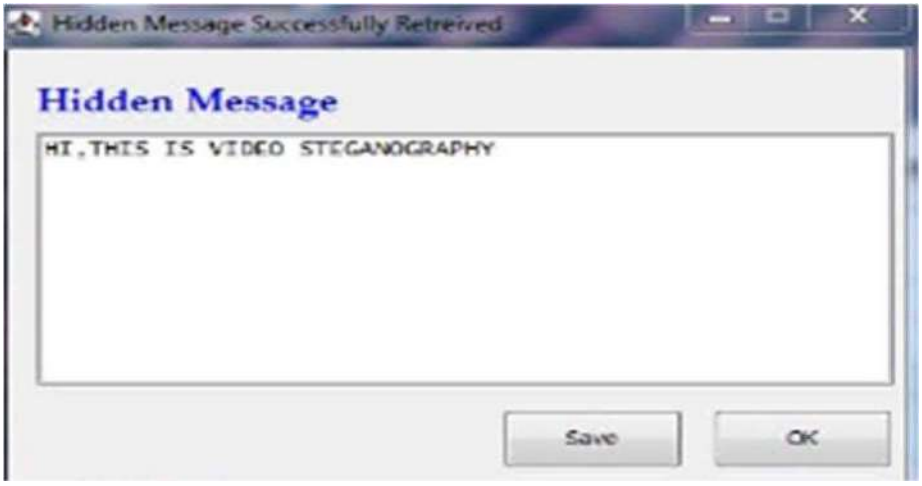


Fig 7.7 Secret message retrieved

Fig 7.7 represents the page of successfully retrieving the secret message with the message displayed.

Fig 7.8 Hiding of Secret message in the form of image

Fig 7.8 represents the page of hiding the secret message in the form of image by giving the input, such as the secret message in the form of image to be hidden and then the cover file as video to embed the secret message and the destination path for the output to be stored and finally the key.

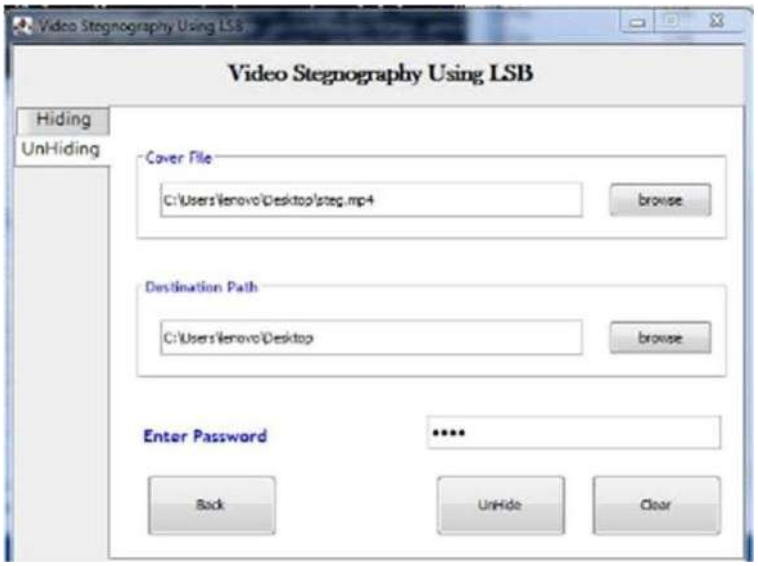


Fig 7.9 Unhiding of Secret message in the form of image

Fig 7.9 represents the page of unhiding the secret message in the form of image by giving the input, such as steg file which is the combination of both the secret message and the cover file, the cover file as video and the destination path for the output to be stored and finally the key.

Fig 7.10: Secret message retrieved

Fig 7.10 represents the page of successfully retrieving the secret message with the image displayed.

7.1 Summary

Project video steganography using LSB technique with encryption using DES, Triple DES, and MD5 is a technique that hides a secret message within a video file. The Least Significant Bit (LSB) technique is used to embed the message within the video file, and DES, Triple DES, or MD5 can be used to encrypt the secret message before it is embedded.

The project involves first encrypting the message using one of the encryption algorithms, and then dividing it into several segments. These segments are then embedded into the video file's LSB of selected frames, making sure that the video's quality is not affected.

To retrieve the hidden message, the video file is read and the LSB of the selected frames is extracted. The extracted bits are then reassembled to form the encrypted message, which is then decrypted using the same encryption algorithm used to encrypt it. The final step is to obtain the original message, which is then revealed to the user.

The project has several advantages, including the ability to hide large amounts of data, security against attacks, and the ability to be used in various fields such as military communication and confidential video data sharing. However, it also has limitations, such

as the possibility of being detected by sophisticated steganalysis techniques and the potential for degradation of the video quality.

Overall, the project video steganography using LSB technique with encryption using DES, Triple DES, and MD5 is an effective way to securely hide confidential messages within video files.

Chapter 8

CONCLUSION AND FUTUREWORK

- The proposed system for video steganography provides dual level security for secret data and is designed based on LSB method. This approach takes less computational time for encryption and decrypting the data which results remarkable growth in performance. This project is an effort to hide our secret data in dual levels in image and text format onto the video.
- The advantages of this approach that it takes less computational time for encryption/decryption of the secret data which results in remarkable growth in performance
- The advantages of this approach that it takes less computational time for encryption/decryption of the secret data which results in remarkable growth in performance.
- When we hide secret data in video there will be no loss in quality of the video and even no one can guess the presence of secret data within the video.
- This project provides a robust and effective solution for secure communication using video steganography, and there is room for further improvement to enhance its capabilities and usability.
- Further this method can be used for real time processing of data since it is very fast in parallelization. This method can be further implemented to hide the secret video in video, such that there will be no loss in quality of the video and even no one can guess the presence of secret data within the video.
- Another potential enhancement is to provide a user-friendly interface for the encryption and decryption process, which would make the project more accessible to non-technical users. Additionally, the project could be extended to support other encryption algorithms for added flexibility
- Future enhancements for this project could include the ability to embed larger messages within the video file, improved encryption algorithms, and the use of multiple video files for embedding the message. Additionally, the project could also be extended to include support for other file formats, such as audio or image files, and the development of a user-friendly interface for ease of use.

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